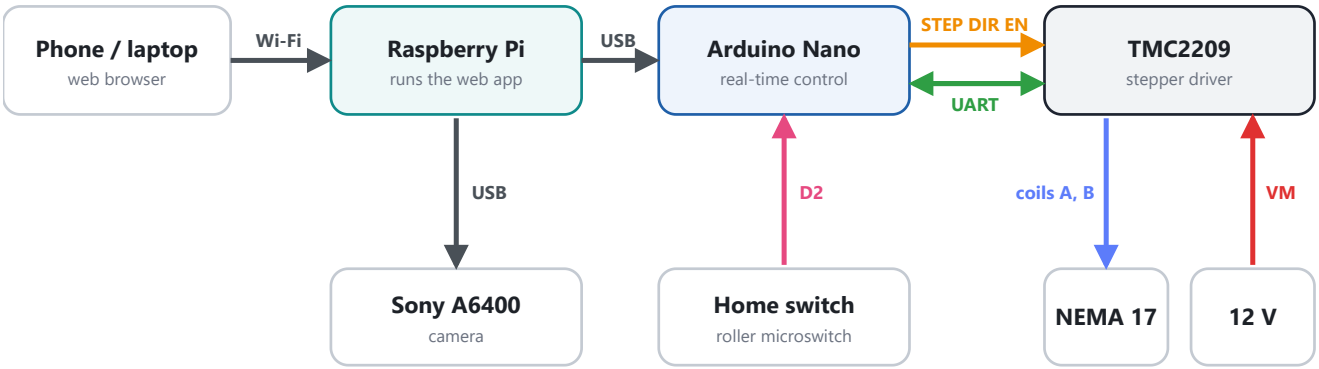
MACRORAIL

Hardware quick start

Wire the Arduino Nano, the TMC2209 driver and the stepper motor on a breadboard, then get the rail moving.

1 Overview

The **Raspberry Pi** runs the web app. It talks to two things over USB: the **Sony camera** (to take the pictures) and an **Arduino Nano**. The Nano does the real-time work: it sends step pulses to the **TMC2209 driver**, which drives the **NEMA 17 motor** from a **12 V supply**, and it watches the **home switch** so the rail always knows where zero is.



What's in this guide

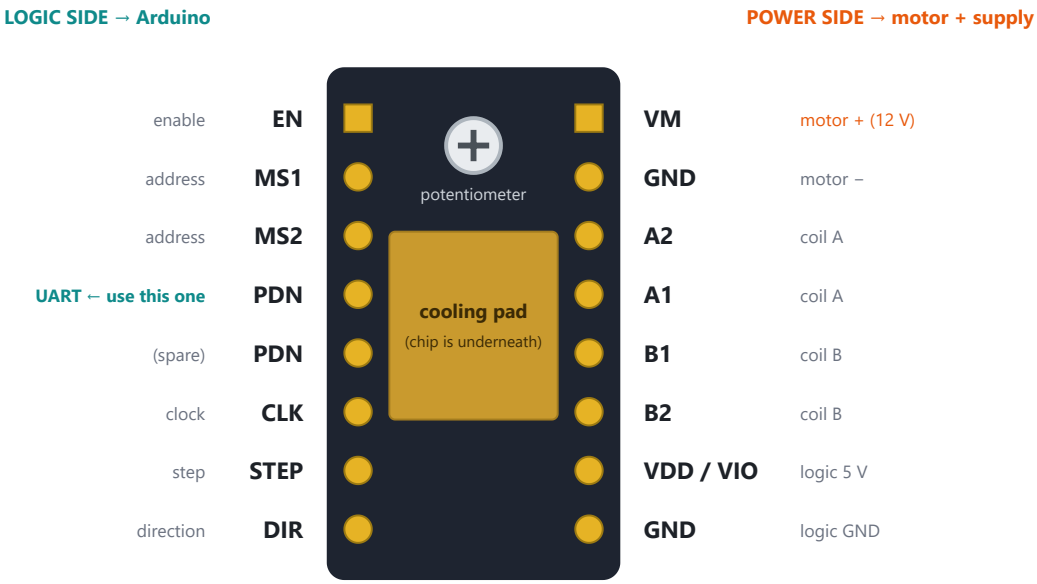
2	**Meet the parts**	The driver, the Nano, resistors, capacitor, switch, and the shopping list.
3	**Build the circuit**	How a breadboard works, the complete wiring picture, and a numbered checklist.
4	**Before you power up**	Coil pairs, the UART pin, the switch and the common ground.
5	**Flash & install**	Firmware onto the Nano, app onto the Pi.
6	**Camera**	A6400 settings.
7	**First power-up checks**	Driver, switch, direction, homing, travel.

> **The one rule that protects the driver**
> Never plug or unplug the motor while the 12 V supply is on. That can destroy the TMC2209.

2 Meet the parts

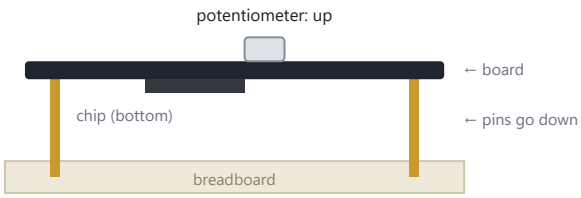
The TMC2209 driver

The driver turns small logic signals from the Nano into the strong currents that move the motor. It has two sides: the **logic side** talks to the Nano, and the **power side** connects to the motor and the 12 V supply.



Top view: potentiometer facing you, pins pointing away (into the breadboard).

The usual TMC2209 layout (BIGTREETECH V1.2/V1.3 and most clones). Check the labels printed on your board's underside.



Side view

Pins go into the breadboard; the potentiometer faces up.

Check your board now

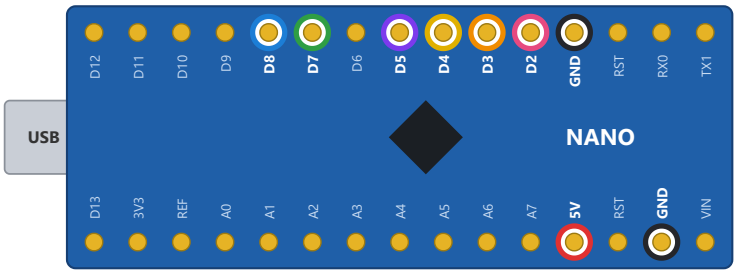
- [ ] Flip it over and find the labels printed next to the pins.
- [ ] Compare them with the picture: EN, MS1, MS2, PDN, PDN, CLK, STEP, DIR on one side; VM, GND, A2, A1, B1, B2, VDD/VIO, GND on the other.
- [ ] Find the **UART pin**: labelled PDN, PDN_UART or UART. On some BTT modules you must bridge a solder jumper to bring it out to the header pin.
- [ ] Names may differ: *OA1/OA2* or *1A/1B* instead of A1/A2, *VIO* or *VCC* instead of VDD.

The potentiometer (the little screw, VREF) sets current the old-fashioned way. MacroRail sets current over UART from the web app, so **leave it alone**.

Sense resistor: usually 0.11 Ω (marked R100 or R110). If yours differs, change `DRIVER_RSENSE_OHM` in `firmware/src/config.h`.

Heatsink? Stick it on the cooling pad on top. It must not touch any pins.

The Arduino Nano



Shown the way it sits on the breadboard: USB on the left.

Pins this project uses

- **D2** home switch (interrupt INT0)
- **D3** STEP
- **D4** DIR
- **D5** EN (enable)
- **D7** UART in (direct)
- **D8** UART out (through 1 kΩ)
- **5V** logic power
- **GND** ground (both GND pins are used)

MacroRail uses six signal pins (D2–D5, D7, D8), plus 5V and both GND pins. The pin names are printed on the Nano next to each pin. D2 is the hardware-interrupt pin, so the home switch stops the motor the instant it closes. D6 is unused and D9 stays free for a future shutter-cable relay. A CH340 clone is fine; it is powered by the Pi's USB.

The same colours are used for every wire in this guide, so you can match picture, checklist and board at a glance.

The resistors and the capacitor

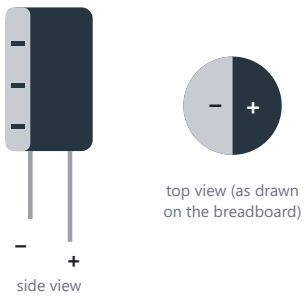


1 kΩ (R2): brown · black · red · gold
5-band: brown · black · black · brown · brown
UART: D8 talks through it



10 kΩ (R1): brown · black · orange · gold
5-band: brown · black · black · red · brown
EN pull-up: keeps the motor off while the Nano boots

Resistors have no direction. Not sure about the colours? Measure: a multimeter on Ω reads about 1.0 k or 10 k.



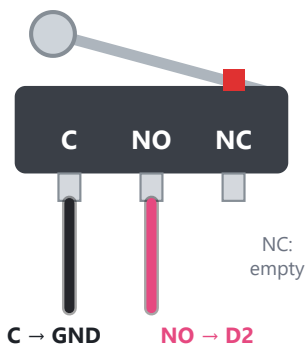
100 µF electrolytic capacitor (C1)

Stripe side = minus (−), longer leg = plus (+).
It goes across VM and GND right next to the driver and soaks up voltage spikes from the motor.

Backwards it can pop, so check twice: − to GND, + to VM.

Rating: 25 V or higher.

The home switch



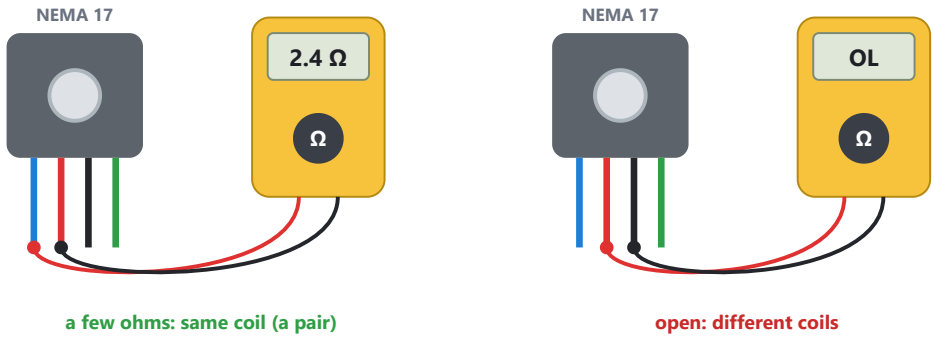
A roller microswitch has three terminals: **C** (common), **NO** (normally open) and **NC** (normally closed). Wire **C and NO**: the switch closes when pressed and pulls D2 to GND. NC stays empty.

D2 is the Nano's hardware-interrupt pin (INT0), so the firmware stops any motion **the instant the switch closes**, in either direction.

If you ever rewire it NC, set `ENDSTOP_ACTIVE_LEVEL` to `HIGH` in `firmware/src/config.h`.

The motor

The NEMA 17 (from the Anet A8) has 4 wires: two coils. With a multimeter on Ω, the two wires of one coil read a few ohms; wires from different coils read open. One pair goes to A1/A2, the other to B1/B2. Wire colours vary between motors, so measure rather than trust them.



Multimeter on Ω: touch two wires at a time until you find the two pairs. Mark them (tape, a pen) before wiring.

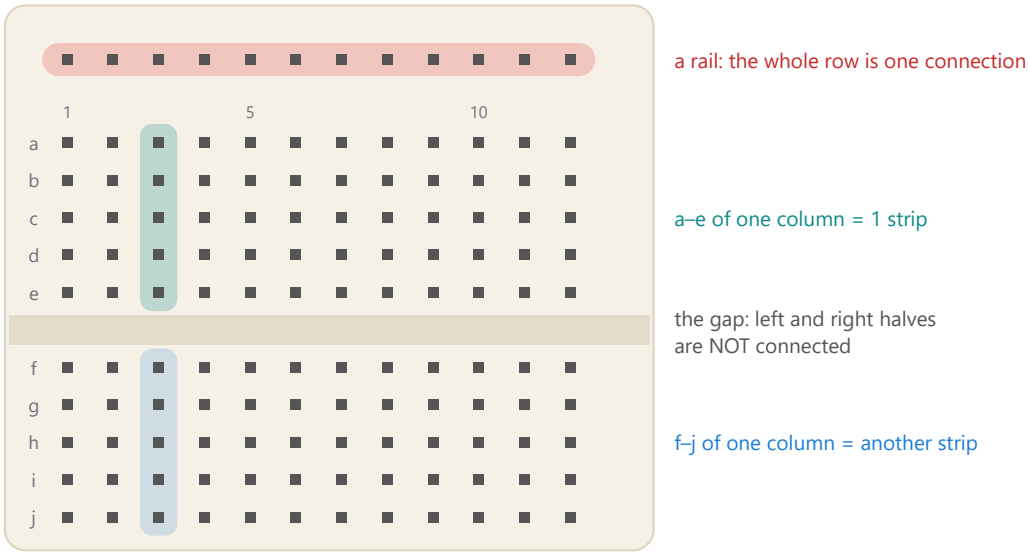
What you need

Part	Notes
Arduino Nano (CH340 clone is fine)	USB to the Pi, which also powers it.
TMC2209 module (BTT / FYSETC style)	Must expose the PDN_UART pin. Sense resistor usually 0.11 Ω (R100/R110).
NEMA 17 stepper (from the Anet A8)	4 wires = two coils.
12 V supply	Powers the motor (VM) only. Its ground is shared with the Nano's.
Roller microswitch	Wired C + NO, to D2.
Resistors: 1 kΩ and 10 kΩ	UART (R2) and EN pull-up (R1).
Capacitor: 100 µF, 25 V or more	Across VM and GND at the driver (C1).
Breadboard + jumper wires	Half-size (30 columns) is enough. Male-male jumpers in several colours.
Raspberry Pi + proper 5.1 V / 2.5 A supply	The Pi powers the Nano and the camera's USB link. Under-voltage causes USB drop-outs.
Sony A6400 + USB cable	Plugs into the Pi.
Multimeter	For coil pairs and the checks in section 4.

3 Build the circuit

The breadboard

Inside a breadboard, metal strips join certain holes together. Knowing which holes are joined is the whole trick:



Each column has two separate 5-hole strips, one above and one below the gap. The rails run the full length.

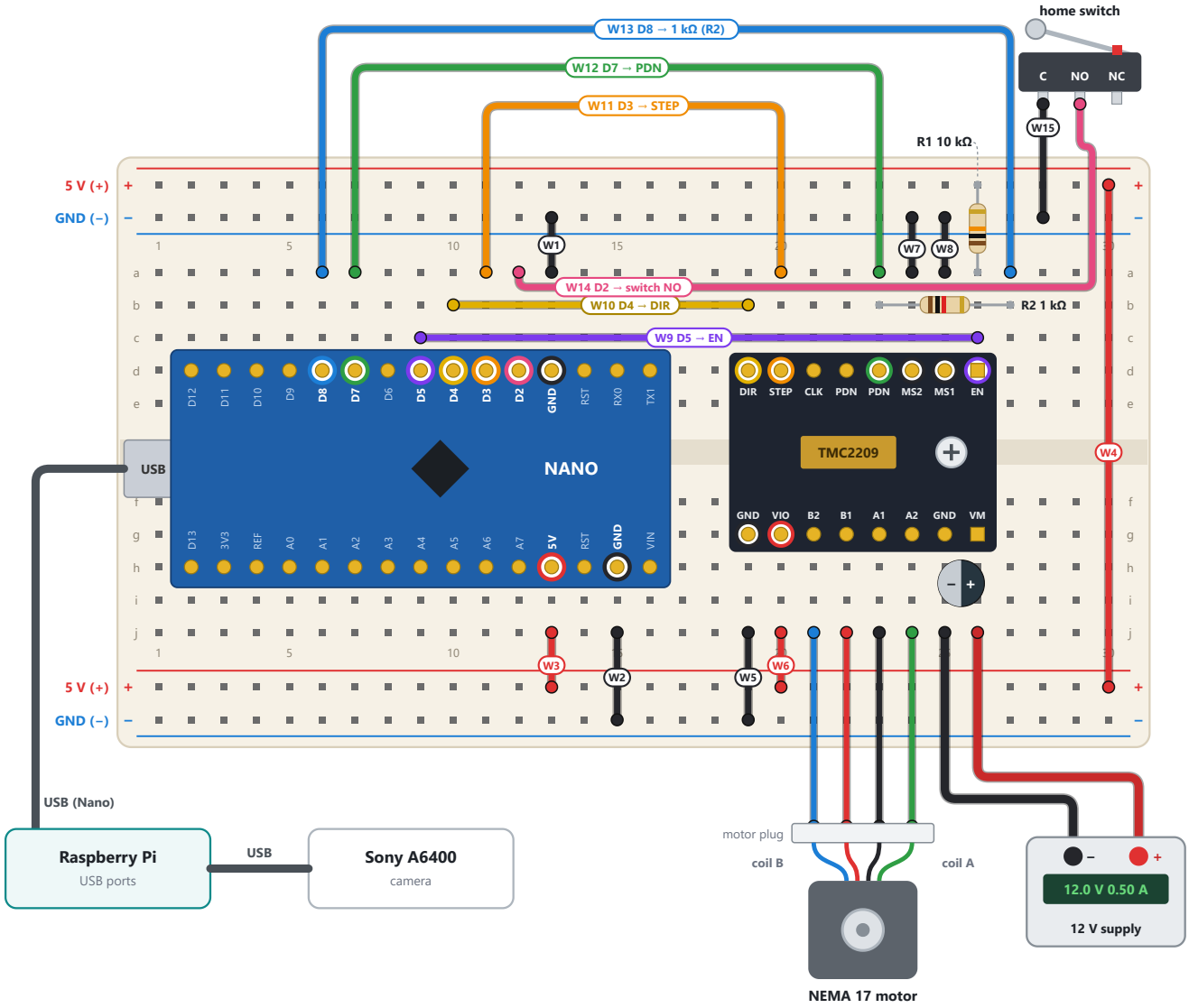
- Columns are numbered (1–30) and rows are lettered (a–j). A hole is named like `c13`: row c, column 13.
- Two parts are connected when their legs are in the **same strip**.
- Chips straddle the gap so that each pin gets its own strip.
- Follow your board's **red +** and **blue −** markings for the rails; their order varies between brands. On a full-size board, long rails are sometimes split in the middle (a break in the red/blue line). This build only uses columns 1–30, so it stays on one side of the split.

How to read the wiring picture

- Every wire has a tag like **W12 D7 → PDN**: its number in the checklist, where it starts and where it ends.
- Most signal wires arch **above** the board so they don't cross. W9 and W10 lie flat on the board in rows c and b, and the switch lead W14 runs flat between rows a and b, underneath the arches.
- Wire colours don't matter electrically, but using the same colours as the pictures makes checking much easier: **black = ground**, **red = 5 V**, and one colour per signal.
- The 12 V leads go **straight** to the driver's VM and GND pins. They never touch the rails, which carry 5 V for the Nano.

> **Before you start building**
> USB unplugged, 12 V supply off and disconnected.

The complete build



Complete wiring. The motor and the 12 V leads go on last, with the supply off (section 7). The TMC2209 is the same board as in section 2, turned a quarter-turn clockwise: logic pins along the top, power pins along the bottom.

Quick reference

Nano	Goes to	Why
D2	Switch NO (switch C → GND)	Closes (LOW) when pressed. Interrupt pin: motion stops instantly.
D3	TMC STEP	One pulse = one 1/16 microstep = 2.5 µm.
D4	TMC DIR	Positive = away from the switch.
D5	TMC EN, plus 10 kΩ to 5 V	Low = motor on. The pull-up keeps the motor off while the Nano boots or is unplugged.
D7	TMC PDN_UART, direct	UART receive.
D8	TMC PDN_UART through 1 kΩ	UART transmit. Single-wire UART needs this resistor.
5V / GND	TMC VIO / GND	Logic supply and common ground.
—	TMC MS1, MS2 → GND	UART address 0.
—	12 V → VM, 12 V − → GND	Motor power. 100 µF across VM–GND at the module.
D9	nothing yet	Reserved for the shutter-cable relay.

Wire it in this order

Tick off one line at a time and compare with the picture on the previous page. “Top rails” are above row a, “bottom rails” below row j. Holes are named row + column, like `c13`.

1 · Push the boards in

☐	**P1**	**Arduino Nano**	pins in rows **d** and **h**, columns **2–16**	D12/D13 in column 2, USB sticking out to the left
☐	**P2**	**TMC2209**	pins in rows **d** and **g**, columns **19–26**	DIR in d19, EN in d26, VM in g26; potentiometer on the right

2 · Wires and parts

☐	**W1**	a13 (Nano GND, top row)	→	top GND rail	
☐	**W2**	j15 (Nano GND, bottom row)	→	bottom GND rail	
☐	**W3**	j13 (Nano 5V)	→	bottom 5 V rail	
☐	**W4**	bottom 5 V rail	→	top 5 V rail	column 30, links the rails
☐	**W5**	j19 (TMC GND)	→	bottom GND rail	logic ground
☐	**W6**	j20 (TMC VIO)	→	bottom 5 V rail	logic power
☐	**W7**	a24 (TMC MS2)	→	top GND rail	UART address 0
☐	**W8**	a25 (TMC MS1)	→	top GND rail	UART address 0
☐	**R1**	a26 (TMC EN)	→	top 5 V rail	10 kΩ pull-up
☐	**W9**	c9 (D5)	→	c26 (EN)	lies flat on the board
☐	**W10**	b10 (D4)	→	b19 (DIR)	lies flat on the board
☐	**W11**	a11 (D3)	→	a20 (STEP)	
☐	**W12**	a7 (D7)	→	a23 (PDN)	UART, direct
☐	**R2**	b23 (PDN)	→	b27	1 kΩ
☐	**W13**	a6 (D8)	→	a27	UART, through R2
☐	**W14**	switch NO	→	a12 (D2)	runs between rows a and b
☐	**W15**	switch C	→	top GND rail	column 28; NC stays empty
☐	**C1**	100 µF: − stripe in h25 (GND)	→	+ in h26 (VM)	polarity!
☐	**U1**	Nano USB	→	Raspberry Pi	
☐	**U2**	camera USB	→	Raspberry Pi	

3 · Last, with the 12 V supply OFF (during first power-up, section 7)

☐	**M1**	motor coil B	→	j21 (B2) + j22 (B1)	12 V supply OFF
☐	**M2**	motor coil A	→	j23 (A1) + j24 (A2)	12 V supply OFF
☐	**V1**	12 V supply −	→	j25 (TMC GND)	straight to the pin, not a rail
☐	**V2**	12 V supply +	→	j26 (TMC VM)	straight to the pin, not a rail

Done with part 2? Hold the board next to the picture and check every wire once more against the quick reference, then continue with section 4.

4 Before you power up

- [ ] **Coil pairs found.** Few ohms between the two wires of a pair, open between pairs. One pair on A1/A2, the other on B1/B2.
- [ ] **UART pin right.** W12 and R2 land on the PDN pin marked *use this one* (PDN / PDN_UART / UART), solder jumper bridged if your module needs it.
- [ ] **Switch on C + NO** (closes when pressed): C to the GND rail, NO to D2, NC empty. The firmware stops any motion the instant the switch closes, in either direction. If you ever rewire it NC, set `ENDSTOP_ACTIVE_LEVEL` to `HIGH` in `firmware/src/config.h`.
- [ ] **Motor direction** is set by `INVERT_DIRECTION` in `firmware/src/config.h`, already set for this rail.
- [ ] **Common ground.** 12 V −, TMC GND and Nano GND must all connect. The TMC's two GND pins are joined on the module: a multimeter on beep between j25 and the bottom GND rail should beep.
- [ ] **No short.** Between the 5 V rail and the GND rail the multimeter must **not** beep.
- [ ] **C1 the right way round:** stripe (−) in h25, + in h26.

> Never plug or unplug the motor while 12 V is on. Ignore the VREF potentiometer: current is set over UART from the web app (Settings → Motor current).

5 Flash & install

Flash the Nano

Once the app is installed (below), plug the Nano into the Pi and flash it there. Stop the app first, because it owns the serial port:

```
uv tool install platformio
sudo systemctl stop macrorail
pio run -d ~/macrorail/firmware -e nano -t upload
```

Or flash from any computer with PlatformIO: `pio run -d firmware -e nano -t upload`.

Use `-e nano_new` if the upload fails with a genuine or new-bootloader Nano.

Install the app on the Pi

On the Pi, clone the repository and run the install script (it asks for your sudo password):

```
git clone https://github.com/Sam-de-Ham/MacroRail.git ~/macrorail
~/macrorail/deploy/install.sh
```

Then find the Nano's port and put it in the config if it differs from the default:

```
ls /dev/serial/by-id/
nano ~/.config/macrorail/config.toml    # [rail] port = "/dev/serial/by-id/..."
sudo systemctl restart macrorail
```

Open **http://pi.local:8000** on your phone or laptop. Logs: `journalctl -u macrorail -f`. Autostart off: `sudo systemctl disable --now macrorail`.

6 Camera

Sony A6400 settings

- Menu → USB Connection: **PC Remote**. Still Img. Save Dest.: **Camera Only** (images stay on the card).
- **Manual focus, manual exposure.** Silent shooting (electronic shutter) avoids shutter shock.
- Turn **auto power-off off**; use a dummy battery or a full one for long stacks.
- Connect the camera's USB to the Pi. The camera pill in the app turns green.

7 First power-up checks

1. **Motor unplugged**, 12 V on, Nano on the Pi. The rail pill shows *Not homed*. *No driver power* means the TMC isn't answering: check 12 V, the PDN_UART wiring (W12, W13, R2) and MS1/MS2 (W7, W8). The TMC2209 only answers on UART when VM has power.
2. **Switch:** press it by hand. The Position card shows *Home switch pressed* immediately.
3. **12 V off**, plug in the motor, 12 V on. Jog **Away** 1 mm: the carriage must move **away from the switch**. If it moves toward it, swap the two wires of one coil (12 V off!) or flip `INVERT_DIRECTION` in `firmware/src/config.h` and reflash.
4. **Home.** The rail seeks the switch at 1 mm/s and stops the instant it closes. It backs off until the switch opens again, plus 0.5 mm, then re-approaches very slowly (0.2 mm/s: it looks almost stationary for a few seconds) and parks just clear of the switch. The activity log shows how far your switch travels between closing and opening. Keep a finger near the 12 V switch the first time.
5. **Measure travel.** Jog to the far end of the usable travel and note the position: that becomes `travel_mm` in the config (default 140).

> **Then:** set a start and end on the live view, preview the range, and run a short test stack.
> Defaults: 600 mA run / 200 mA hold, 2 mm/s travel, 0.8 s settle, 10 µm step.